

AN8280G-TK Power Supply Replacement Procedure

PSU REPLACEMENT

Step 1: Preparation

- Turn AC power OFF at service panel.
- Remove front and rear doors.
- Turn power OFF at PSU.
- Verify all LEDs are OFF on front of PSU.
- Wait 5 minutes.

LOCKOUT/TAGOUT PROCEDURE

This material is to be applied to all replacement procedures that involve the Scaleable Gradient Cabinet modules.

DANGER!!

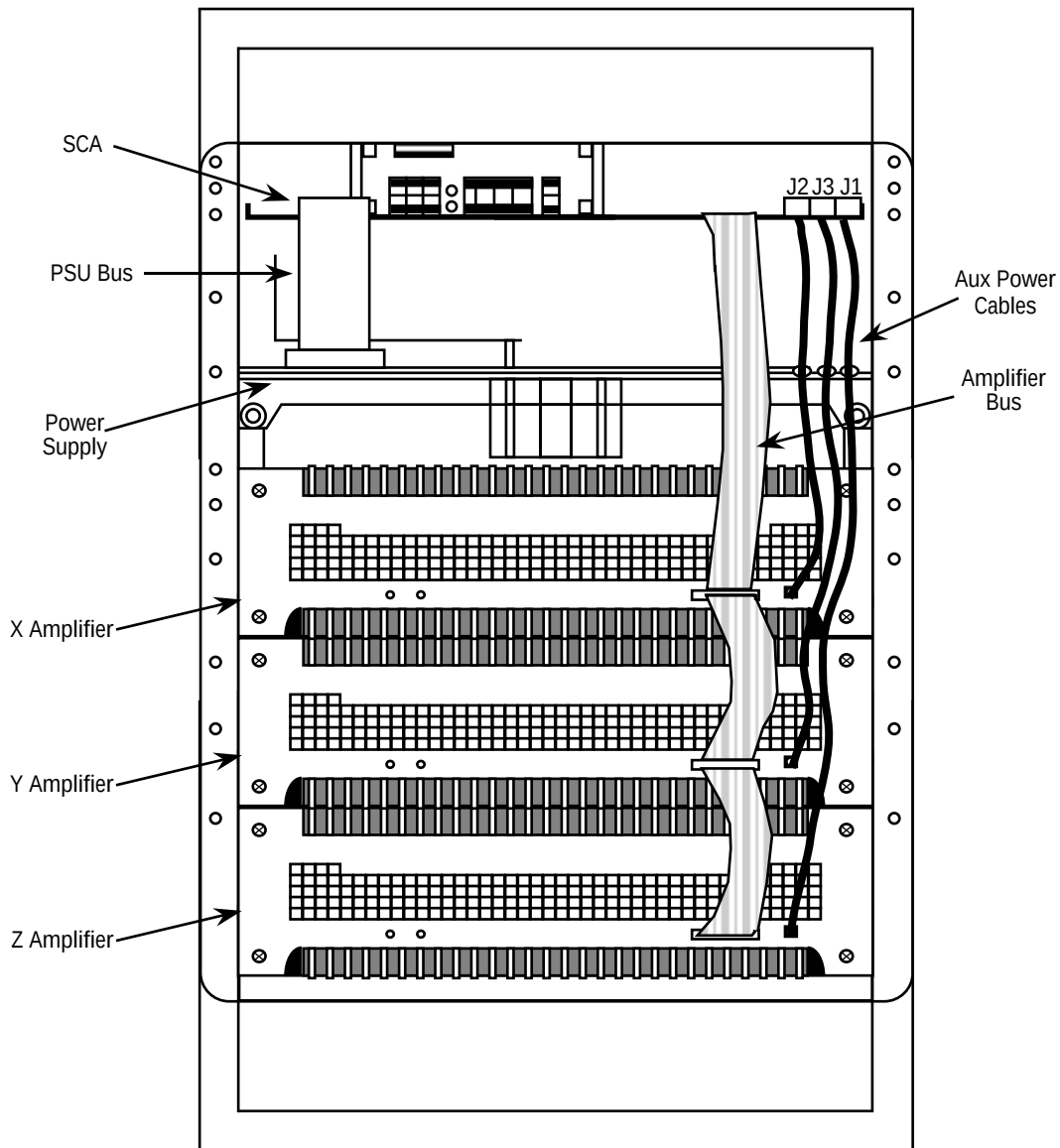
FATAL ELECTRIC SHOCK HAZARD! TO PREVENT FATAL ELECTRIC SHOCK, DISCONNECT POWER FROM THE PDU BEFORE YOU PERFORM THE FOLLOWING PROCEDURES. PERFORM LOCKOUT/TAGOUT PROCEDURE PER GE OSHA LOCKOUT/TAGOUT REQUIREMENTS 29 CFR 1910.147. DO THIS BY SECURING THE PDU CIRCUIT BREAKER FOR THE SCALEABLE GRADIENT CABINET.

1. Turn off the SGD Cabinet Breaker at the Power Distribution Unit. Lock out the Breaker and tag it.
2. After power to the SGD Cabinet has had sufficient time to dissipate, take a Digital Multimeter and set it to its highest AC voltage range.
3. Verify that all energy has been dissipated by measuring incoming power to all components of the Scaleable Gradient Cabinet.
 - Place the reference probe (black) on the SGD Cabinet Ground.
 - Locate L1, L2, and L3. These are the 208V, 3 Phase inputs to the SGD Base SR-20 Cabinet.
4. Measure voltage at each of three 208 volt input terminals. The meter should read 0 volts AC at each of the three measuring points. See below.

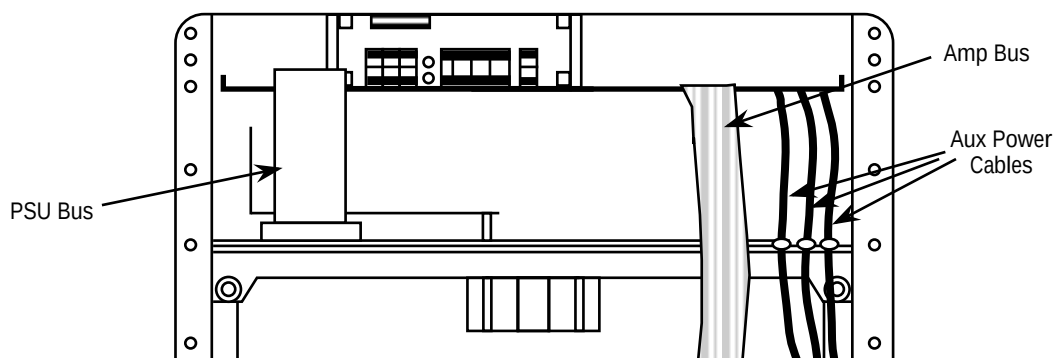
Measure Incoming
Power Here
(L1, L2, L3)

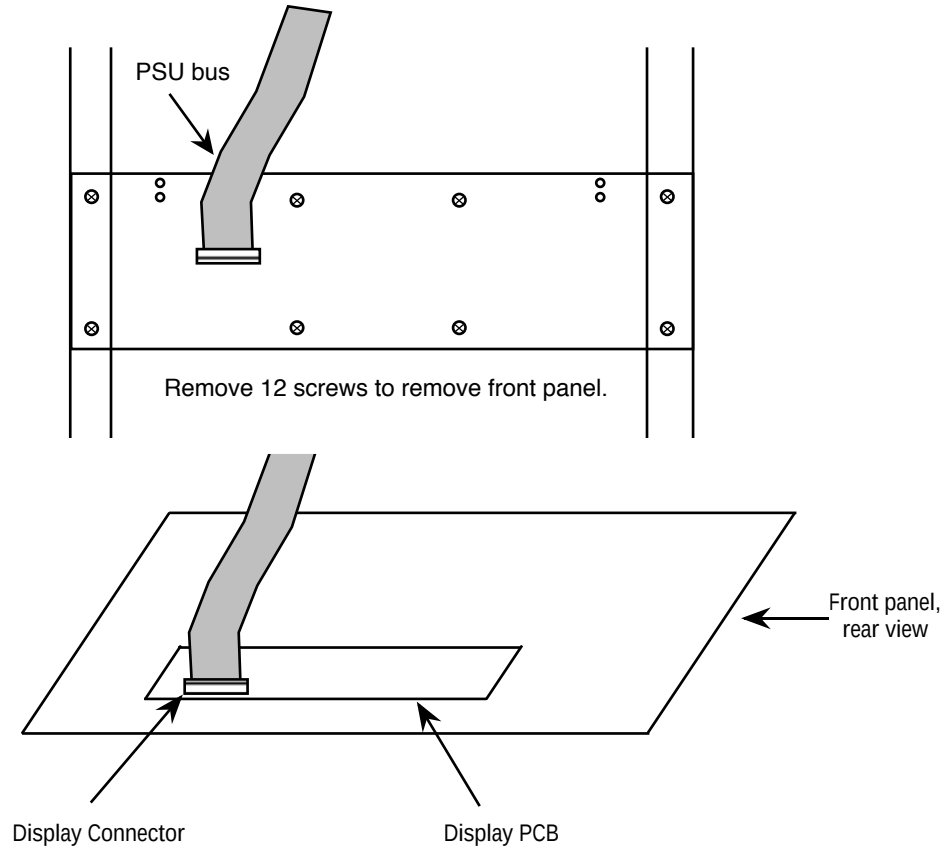
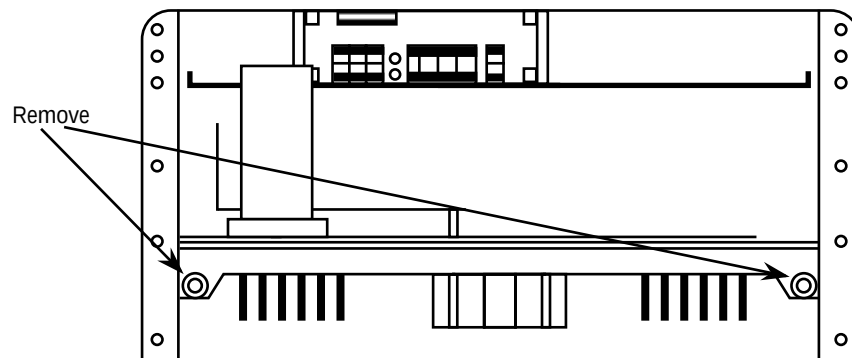


Step 2: Disconnect Auxiliary Power Cables (3) from Front of SCA. Disconnect Amplifier Bus Cable from all 3 Amps and Drape Carefully to Avoid Damage.

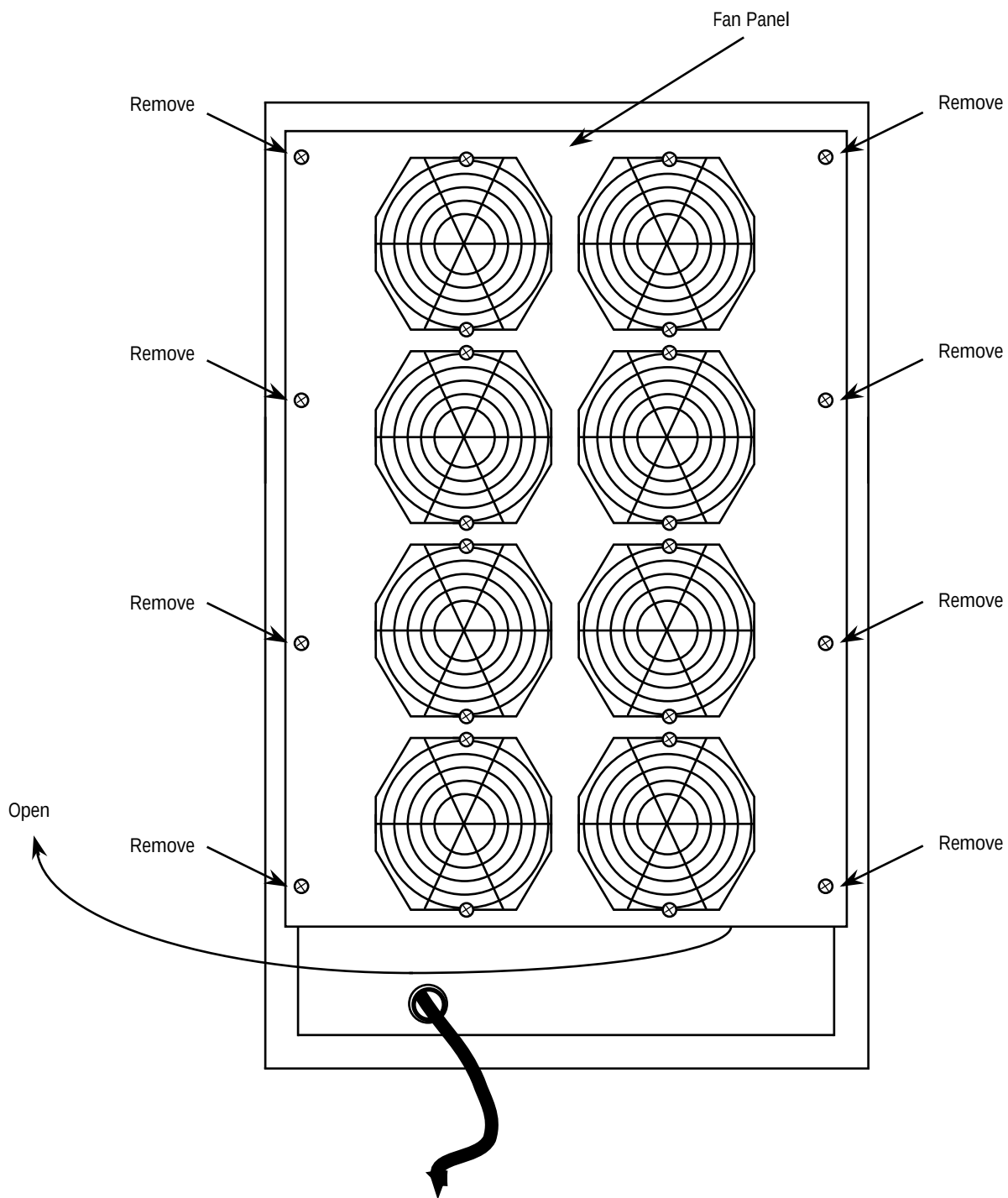


Step 3: Disconnect PSU Bus at Front of PSU, Amp Bus and Aux Power Cables to Amplifiers and Drape Over Top of Rack.

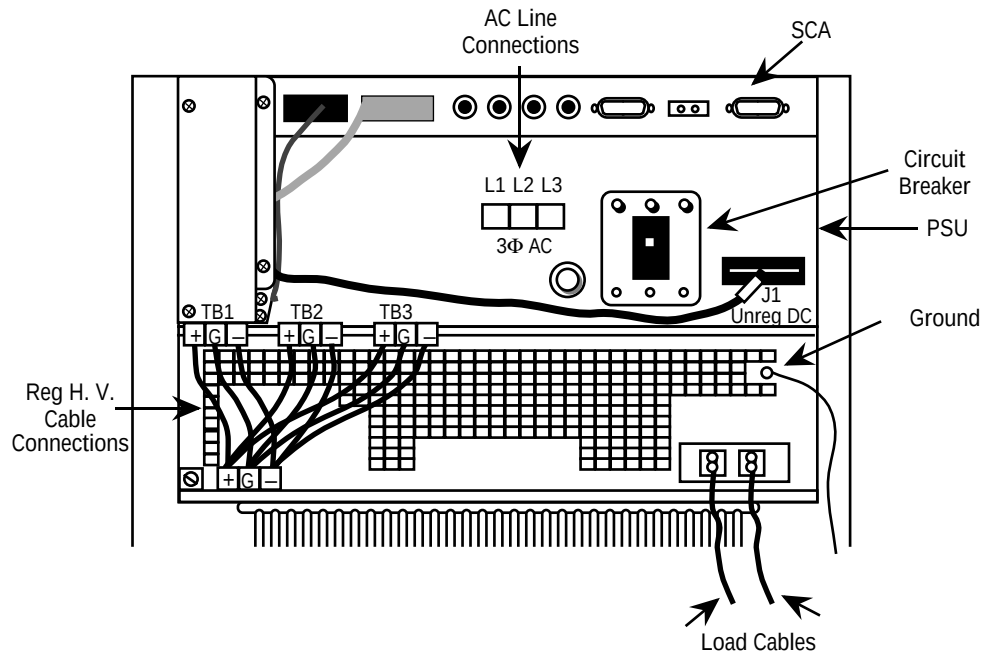


Step 4: Remove PSU/SCA Common Front Panel and Disconnect Display Connector at Display Card.**Step 5: Loosen Captive Screws (2) at Front, Holding PSU to Sleeve.**

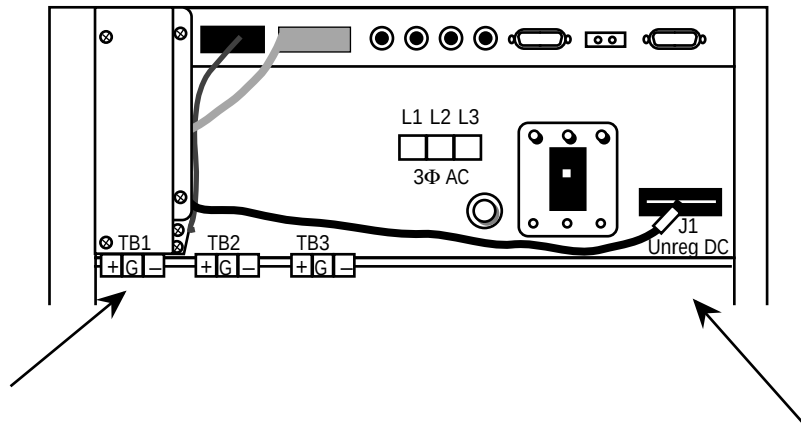
Step 6: At Rear of Rack, Remove Screws (8) and Open Fan Panel.



Step 7: Disconnect Reg H.V. DC Cables (3), Unreg H.V. DC and Remove 200 VAC Line from Rear of PSU. Use Offset Screwdriver Provided in Rear of Rack.

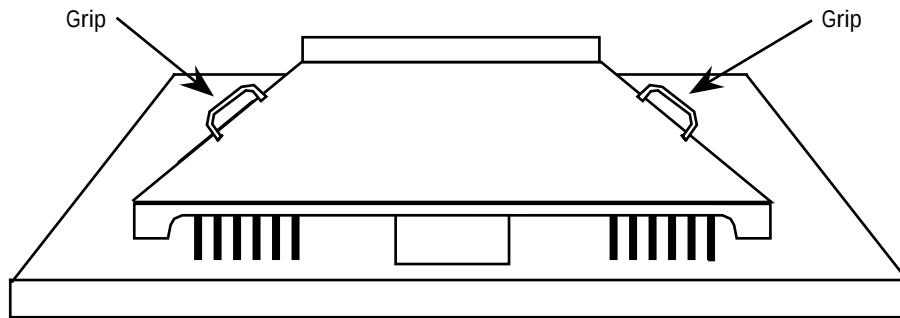


Step 8: Push PSU Forward in Rack 5 cm by Pushing on Rear of PSU.

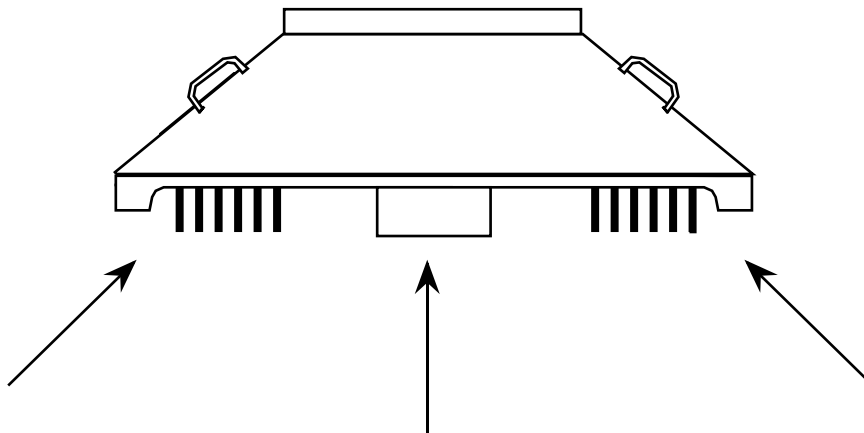


Step 9: Remove PSU from Front.

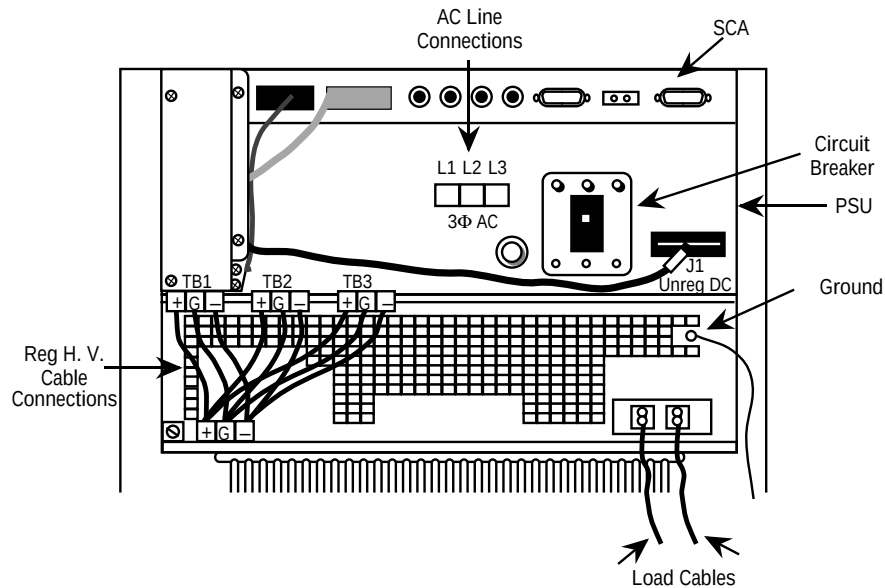
- Carry by handles. Be careful to avoid damage to PSU.



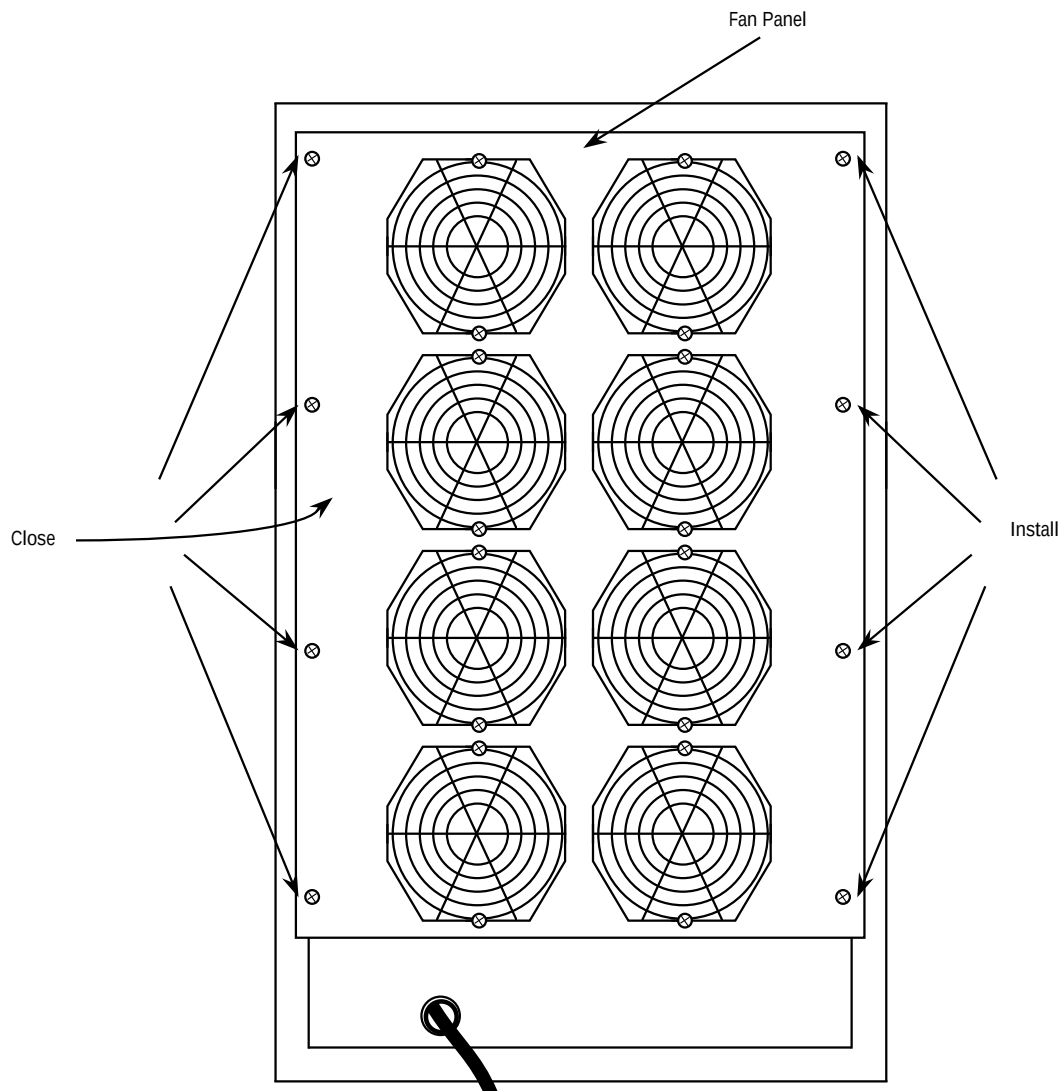
Step 10: Place New PSU in Sleeve and Slide in from Front to Back. Engage Alignment Pins.



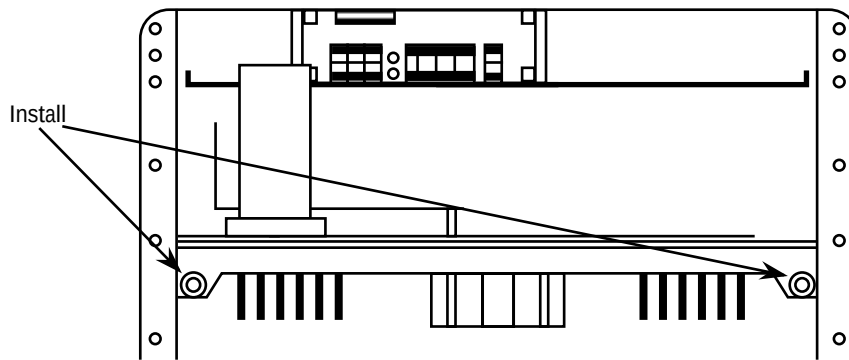
Step 11: Install Cables in PSU Terminals and Connectors. Use Offset Screwdriver Provided. Turn Circuit Breaker On.



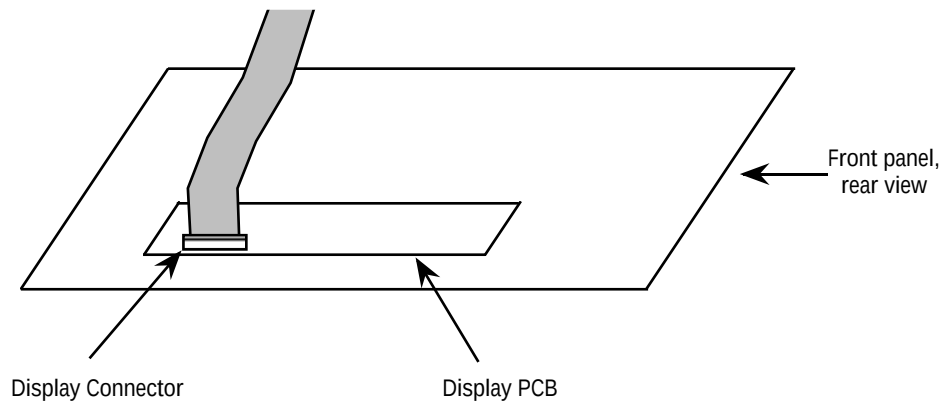
Step 12: Close Fan Panel and Install 8 Screws.

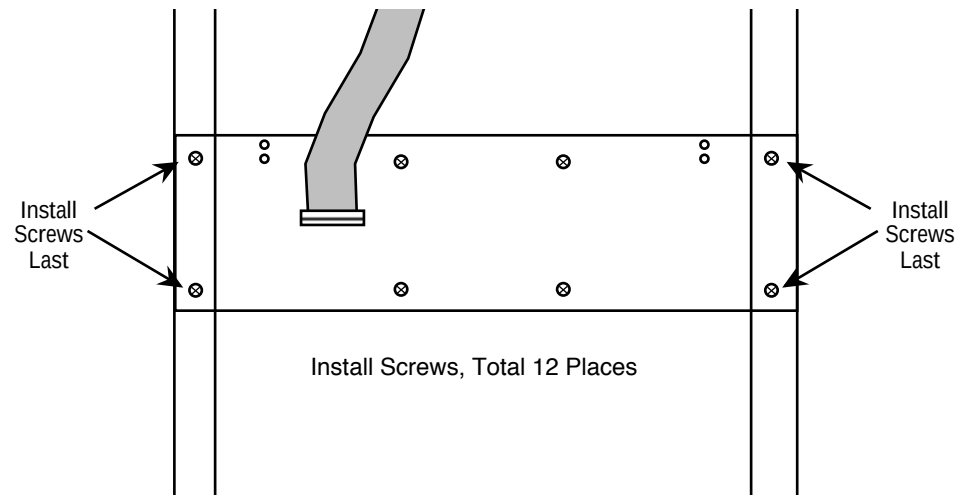


Step 13: Install Front Panel Screws Holding PSU to Sleeve.

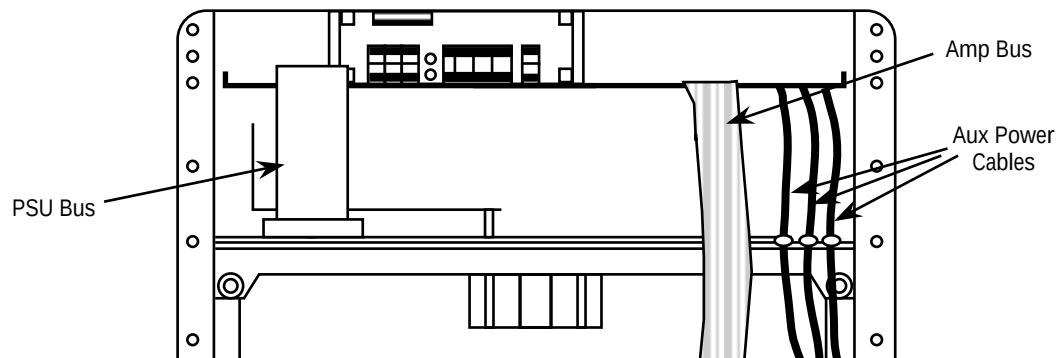


Step 14: Install Display Connector on Rear of Front Panel Display PCB, Observing Polarization Key. Then Install Front Panel of PSU/SCA.

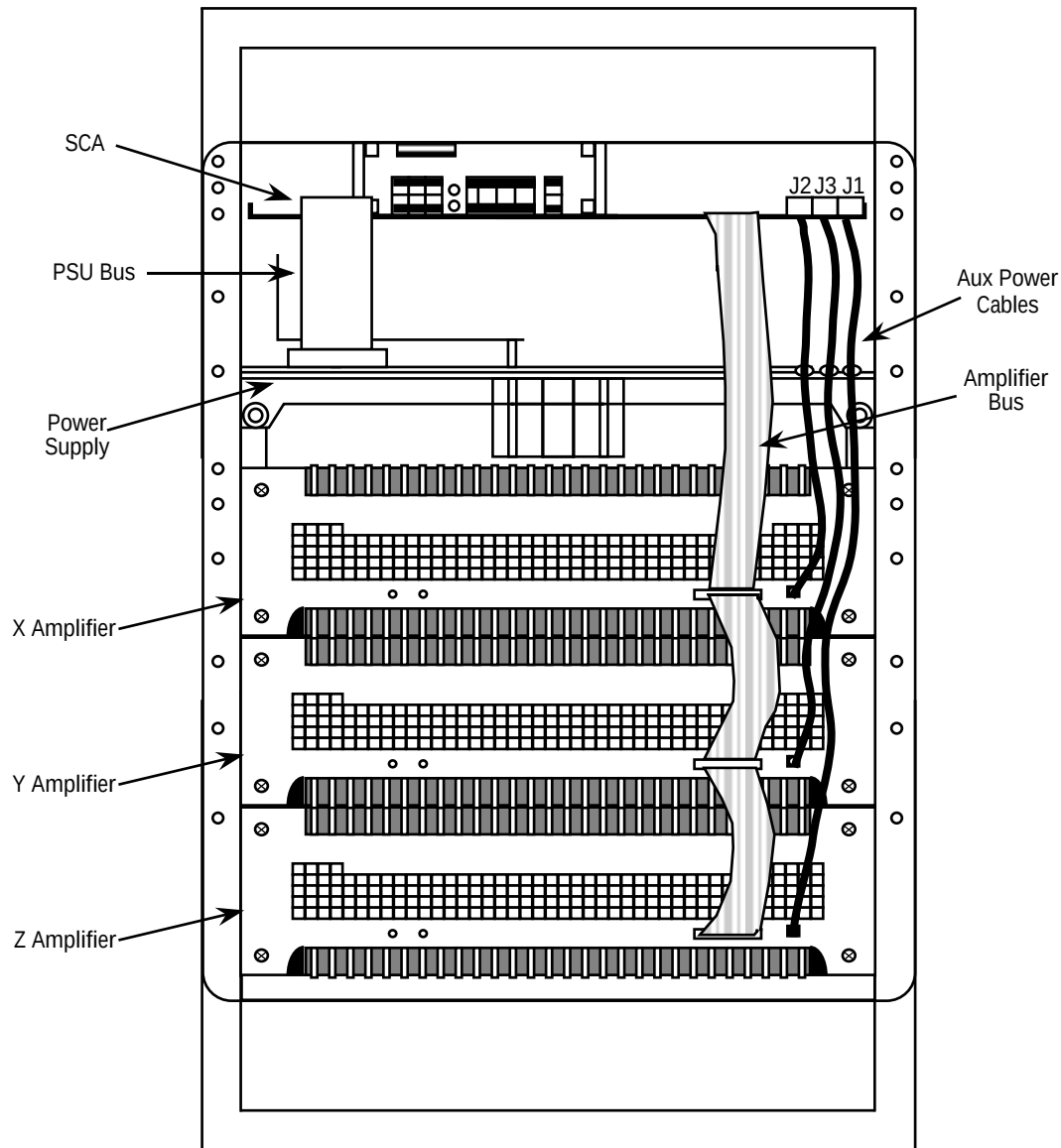


**Step 15: Install PSU Bus Cable.**

- Reconnect Amp Bus cable and Aux Power cables to all amplifiers. Be sure connectors are latched.



Step 16: Connect Aux Power Cables (3) on Front of SCA. Connect Amp Bus Cable to all 3 Amps.



Step 17: Inspect Unit Prior to Turn-on.

- Install front and rear doors.
- Turn AC power ON at service panel.
- Verify proper power-up sequence.

FUNCTIONAL CHECKS REQUIRED

No calibrations required; power up the cabinet and perform a scan to verify functionality.